

sPHENIX Simulation Tools

Pythia 8 · HIJING · Geant4 — 10-Week Beginner Course

Course Overview

This 10-week (~2.5 month) course introduces the three pillars of heavy-ion and collider Monte Carlo workflow: **Pythia 8** for p+p event generation, **HIJING** for heavy-ion event generation (A+A), and **Geant4** for full detector simulation. The course is designed for beginners with basic C++/Linux familiarity, and can be completed either in the sPHENIX Singularity container at BNL/SDCC or on a personal Linux/macOS/WSL2 machine.

By the end you will be able to: (1) generate minimum-bias and hard-QCD p+p events, (2) produce Au+Au heavy-ion events comparable to sPHENIX physics runs, (3) propagate those particles through a simplified sPHENIX calorimeter stack in Geant4, and (4) analyze the resulting hits with ROOT.

Prerequisites

- Basic C++ (classes, pointers, Makefiles) — you do not need to be an expert.
- Comfort with a Linux shell (cd, ls, scp, tar, export).
- ROOT ≥ 6.22 or willingness to install it (covered in Week 1).
- ~15 GB free disk, ~4 GB RAM available to a VM/WSL2.

Software Stack

Tool	Version	Purpose
Pythia8	8.310+	p+p event generation
HIJING	1.383 / 2.0	Heavy-ion A+A event generation
Geant4	11.2.x	Detector simulation
ROOT	6.30+	Analysis & I/O
HepMC3	3.2.7+	Event record format
LHAPDF6	6.5+	Parton distributions (optional)
Fun4All	sPHENIX build	sPHENIX framework (optional, for Module 8)

Weekly Schedule

Wk	Module	Topic	Deliverable
1	M1	Intro, environment & sPHENIX overview	Working Geant4/Pythia/ROOT install
2	M2	Pythia 8 basics: event record, pp minbias	100k pp minbias events, pT spectrum
3	M3	Pythia 8 advanced: hard QCD, tunes, LHAPDF	Jet pT spectrum at 200 GeV
4	M4	HIJING theory: soft+hard, nuclear geometry	Written summary of mini-jet model

5	M5	HIJING hands-on: build & run AuAu 200 GeV	$dN/d\eta$; plot vs. centrality
6	M6	Geant4 basics: geometry, materials, physics lists	Single-particle gun simulation
7	M7	Geant4 sensitive detectors & hit readout	Sampling calorimeter w/ ROOT tree
8	M8	sPHENIX / Fun4All framework introduction	Run Fun4All_G4_sPHENIX tutorial macro
9	M9	Integration: generator $\rightarrow$ Geant4 $\rightarrow$ analysis	Completed practice example
10	—	Final project & review	Mini-report + plots

Reading List (Primary)

- Sjöstrand et al., *An introduction to PYTHIA 8.3*, SciPost Phys. Codebases 8 (2022). arXiv:2203.11601.
- Wang & Gyulassy, *HIJING: A Monte Carlo model for multiple jet production in pp, pA and AA collisions*, Phys. Rev. D 44, 3501 (1991).
- Agostinelli et al. (Geant4 Collaboration), *Geant4 — a simulation toolkit*, NIM A 506 (2003) 250.
- Allison et al., *Recent developments in Geant4*, NIM A 835 (2016) 186.
- sPHENIX Technical Design Report, BNL-PUB-2024-001 (latest revision).
- Geant4 Book For Application Developers, current release (HTML).

Grading / Self-Assessment

Since this is a self-study course, each module ends with a short self-assessment checklist. Aim to satisfy every checkbox before moving on. The final practice example in Week 9–10 is the capstone.

Time Commitment

Plan on about **8–10 hours per week**: ~2 h reading, ~2 h lecture notes, ~4–6 h hands-on code. Heavier weeks (6, 7, 9) may require more lab time.